ON-BOARD DIAGNOSTIC [TWO-STEP DEPLOYMENT CONTROL SYSTEM]

id0802b2001500

Outline

- The air bag system has an on-board diagnostic function to facilitate the system diagnosis.
- The on-board diagnostic function consists of the following functions:
  - — A malfunction detection function, which detects overall malfunctions in the air bag system-related parts.
  - — A memory function, which stores detected DTCs.
  - — A display function, which indicates malfunction locations and status via DTC output.
  - — A PID/data monitoring function, which reads out specific input/output signals and verifies the input/output condition.
- Using the Mazda Modular Diagnostic System (M-MDS), DTCs can be read out and deleted, and the PID/data monitoring function can be activated.

Function

Self-diagnostic function

Memory function

- Stores malfunctions detected in the air bag system-related parts using the malfunction detection function, and the stored malfunction contents are not cleared even if the ignition is switched off or the negative battery cable is disconnected.

Display function

- When the malfunction detection function detects a malfunction, the air bag system warning light illuminates to advise the driver. Using the external tester communication function, DTCs can be output to the DLC-2 via the CAN.

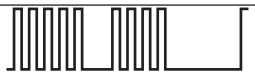
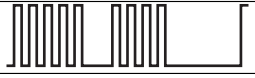
X: Applicable
−: Not applicable

DTC				System malfunction location	Fail-safe function	Drive cycle	Self test type* 1	Memory function
M-MDS display	Air bag system warning light							
	Flashing pattern		Priority ranking					
B0001:11	19		15	Driver-side air bag module (inflator No.1) circuit short to body ground	—	—	C, D	×
B0001:12				Driver-side air bag module (inflator No.1) circuit short to power supply	—	—	C, D	×
B0001:13				Driver-side air bag module (inflator No.1) circuit open circuit or resistance high	—	—	C, D	×
B0001:19				Short circuit to driver-side air bag module (inflator No.1) and other air bag module circuits	—	—	C, D	×
B0001:1A				Driver-side air bag module (inflator No.1) circuit resistance low	—	—	C, D	×
B0001:55	54		6	Configuration setting error (driver-side air bag module (inflator No.1) structural malfunction)	—	—	C, D	×
B0002:11	19		15	Driver-side air bag module (inflator No.2) circuit short to body ground	—	—	C, D	×
B0002:12				Driver-side air bag module (inflator No.2) circuit short to power supply	—	—	C, D	×
B0002:13				Driver-side air bag module (inflator No.2) circuit open circuit or resistance high	—	—	C, D	×
B0002:19				Short circuit to driver-side air bag module (inflator No.2) and other air bag module circuits	—	—	C, D	×
B0002:1A				Driver-side air bag module (inflator No.2) circuit resistance low	—	—	C, D	×
B0002:55	54		6	Configuration setting error (driver-side air bag module (inflator No.2) structural malfunction)	—	—	C, D	×
B0003:53	—	Flashing	4	Operation (deployment) inhibited	—	—	C, D	×

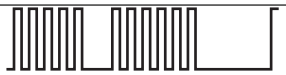
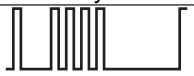
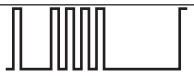
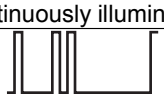
DTC				System malfunction location	Fail-safe function	Drive cycle	Self test type* 1	Memory function
M-MDS display	Air bag system warning light							
	Flashing pattern		Priority ranking					
B0010:11	21		14	Passenger-side air bag module circuit short to body ground	—	—	C, D	×
B0010:12				Passenger-side air bag module circuit short to power supply	—	—	C, D	×
B0010:13				Passenger-side air bag module circuit open circuit or resistance high	—	—	C, D	×
B0010:19				Short circuit to passenger-side air bag module and other air bag module circuits	—	—	C, D	×
B0010:1A				Passenger-side air bag module (inflator circuit resistance low	—	—	C, D	×
B0010:55	54		6	Configuration setting error (passenger-side air bag module (inflator structural malfunction)	—	—	C, D	×
B0050:55	54		6	Configuration setting error (driver-side buckle switch (structural malfunction)	—	—	C, D	×
B0052:55				Configuration setting error (passenger-side buckle switch (structural malfunction)	—	—	C, D	×
B0094:11	42		9	Crash zone sensor circuit short to body ground	—	—	C, D	×
B0094:12				Crash zone sensor circuit short to power supply	—	—	C, D	×
B0094:29				Crash zone sensor internal malfunction	—	—	C, D	×
B0094:2B				Crash zone sensor circuit short to other air bag sensor circuit	—	—	C, D	×
B0094:4A				Crash zone sensor ID error	—	—	C, D	×
B0094:87				Signal reception error from crash zone sensor	—	—	C, D	×
B0094:96				Crash zone sensor internal malfunction	—	—	C, D	×
B0094:55	54		6	Configuration setting error (crash zone sensor structural malfunction)	—	—	C, D	×
B00B5:11	49		26	Seat track position sensor circuit short to body ground	—	—	C, D	×
B00B5:12				Seat track position sensor circuit short to power supply	—	—	C, D	×
B00B5:13				Seat track position sensor circuit open	—	—	C, D	×
B00B5:1D				Seat track position sensor circuit out of range	—	—	C, D	×
B00B5:2B				Short circuit to seat track position sensor and other air bag sensor circuits	—	—	C, D	×
B00B5:55	54		6	Configuration setting error (seat track position sensor structural malfunction)	—	—	C, D	×
B00D2:01	—	—	—	Air bag system warning light malfunction	—	—	C, D	×
B00D2:29	—	—	—	Invalid signal to air bag system warning light	—	—	C, D	×
B00D5:01	18		25	PAD indicator circuit open circuit or short to power supply	—	—	C, D	×
B00D5:29				PAD indicator communication error	—	—	C, D	×
B00D5:55				Configuration setting error (indicator unit structural malfunction)	—	—	C, D	×
B1011:95	58		7	SAS control module connectors are poorly connected	—	—	C, D	×

DTC				System malfunction location	Fail-safe function	Drive cycle	Self test type* 1	Memory function
M-MDS display	Air bag system warning light							
	Flashing pattern		Priority ranking					
B10FD:11	43		13	Driver-side side air bag sensor circuit short to body ground	—	—	C, D	×
B10FD:12				Driver-side side air bag sensor circuit short to power supply	—	—	C, D	×
B10FD:29				Driver-side side air bag sensor internal malfunction	—	—	C, D	×
B10FD:2B				Driver-side side air bag sensor circuit short to other air bag sensor circuit	—	—	C, D	×
B10FD:4A				Driver-side side air bag sensor ID error	—	—	C, D	×
B10FD:87				Signal reception error from driver-side side air bag sensor	—	—	C, D	×
B10FD:96				Driver-side side air bag sensor internal malfunction	—	—	C, D	×
B10FD:55	54		6	Configuration setting error (driver-side side air bag sensor structural malfunction)	—	—	C, D	×
B10FE:11	44		12	Passenger-side side air bag sensor circuit short to body ground	—	—	C, D	×
B10FE:12				Passenger-side side air bag sensor circuit short to power supply	—	—	C, D	×
B10FE:29				Passenger-side side air bag sensor internal malfunction	—	—	C, D	×
B10FE:2B				Passenger-side side air bag sensor circuit short to other air bag sensor circuit	—	—	C, D	×
B10FE:4A				Signal reception error from passenger-side side air bag sensor	—	—	C, D	×
B10FE:87				Signal reception error from passenger-side side air bag sensor ID error	—	—	C, D	×
B10FE:96				Passenger-side side air bag sensor internal malfunction	—	—	C, D	×
B10FE:55	54		6	Configuration setting error (passenger-side side air bag sensor structural malfunction)	—	—	C, D	×
B1126:11	22		20	Driver-side side air bag module circuit short to body ground	—	—	C, D	×
B1126:12				Driver-side side air bag module circuit short to power supply	—	—	C, D	×
B1126:13				Driver-side side air bag module circuit open circuit or resistance high	—	—	C, D	×
B1126:19				Short circuit to driver-side side air bag module and other air bag module circuits	—	—	C, D	×
B1126:1A				Driver-side side air bag module circuit resistance low	—	—	C, D	×
B1126:55	54		6	Configuration setting error (driver-side side air bag module structural malfunction)	—	—	C, D	×
B1127:11	23		19	Passenger-side side air bag module circuit short to body ground	—	—	C, D	×
B1127:12				Passenger-side side air bag module circuit short to power supply	—	—	C, D	×
B1127:13				Passenger-side side air bag module circuit open circuit or resistance high	—	—	C, D	×
B1127:19				Short circuit to passenger-side side air bag module and other air bag module circuits	—	—	C, D	×
B1127:1A				Passenger-side side air bag module circuit resistance low	—	—	C, D	×

DTC				System malfunction location	Fail-safe function	Drive cycle	Self test type* 1	Memory function
M-MDS display	Air bag system warning light							
	Flashing pattern		Priority ranking					
B1127:55	54		6	Configuration setting error (passenger-side side air bag module structural malfunction)	—	—	C, D	×
B1128:11	24		22	Driver-side curtain air bag module circuit short to body ground	—	—	C, D	×
B1128:12				Driver-side curtain air bag module circuit short to power supply	—	—	C, D	×
B1128:13				Driver-side curtain air bag module circuit open circuit or resistance high	—	—	C, D	×
B1128:19				Short circuit to driver-side curtain air bag module and other air bag module circuits	—	—	C, D	×
B1128:1A				Driver-side curtain air bag module circuit resistance low	—	—	C, D	×
B1128:55	54		6	Configuration setting error (driver-side curtain air bag module structural malfunction)	—	—	C, D	×
B1129:11	25		21	Passenger-side curtain air bag module circuit short to body ground	—	—	C, D	×
B1129:12				Passenger-side curtain air bag module circuit short to power supply	—	—	C, D	×
B1129:13				Passenger-side curtain air bag module circuit open circuit or resistance high	—	—	C, D	×
B1129:19				Short circuit to passenger-side curtain air bag module and other air bag module circuits	—	—	C, D	×
B1129:1A				Passenger-side curtain air bag module circuit resistance low	—	—	C, D	×
B1129:55	54		6	Configuration setting error (passenger-side curtain air bag module structural malfunction)	—	—	C, D	×
B1193:00	13		3	SAS control module operation (deployment)	—	—	C, D	×
B1206:00	—	—	—	SAS control module operation (deployment) (fuel cut signal output)	—	—	C, D	×
B1211:11	38		18	Driver-side pre-tensioner seat belt circuit short to body ground	—	—	C, D	×
B1211:12				Driver-side pre-tensioner seat belt circuit short to power supply	—	—	C, D	×
B1211:13				Driver-side pre-tensioner seat belt circuit open circuit or resistance high	—	—	C, D	×
B1211:19				Short circuit to driver-side pre-tensioner seat belt and other air bag module circuits	—	—	C, D	×
B1211:1A				Driver-side pre-tensioner seat belt circuit resistance low	—	—	C, D	×
B1211:55	54		6	Configuration setting error (driver-side pre-tensioner seat belt structural malfunction)	—	—	C, D	×

DTC				System malfunction location	Fail-safe function	Drive cycle	Self test type* 1	Memory function
M-MDS display	Air bag system warning light							
	Flashing pattern		Priority ranking					
B1214:11	39		17	Passenger-side pre-tensioner seat belt circuit short to body ground	—	—	C, D	×
B1214:12				Passenger-side pre-tensioner seat belt circuit short to power supply	—	—	C, D	×
B1214:13				Passenger-side pre-tensioner seat belt circuit open circuit or resistance high	—	—	C, D	×
B1214:19				Short circuit to passenger-side pre-tensioner seat belt and other air bag module circuits	—	—	C, D	×
B1214:1A				Passenger-side pre-tensioner seat belt circuit resistance low	—	—	C, D	×
B1214:55	54		6	Configuration setting error (passenger-side pre-tensioner seat belt structural malfunction)	—	—	C, D	×
B1226:02	36		11	Signal reception error from driver or passenger-side pressure sensor	—	—	C, D	×
B1226:29				Driver-side pressure sensor internal malfunction	—	—	C, D	×
B1226:4A				Driver-side pressure sensor ID mismatch	—	—	C, D	×
B1226:87				Driver-side pressure sensor circuit open	—	—	C, D	×
B1226:96				Driver-side pressure sensor internal malfunction	—	—	C, D	×
B1226:55	54		6	Configuration setting error (driver-side pressure sensor structural malfunction)	—	—	C, D	×
B1227:29	37		10	Passenger-side pressure sensor internal malfunction	—	—	C, D	×
B1227:4A				Passenger-side pressure sensor ID mismatch	—	—	C, D	×
B1227:87				Passenger-side pressure sensor circuit open	—	—	C, D	×
B1227:96				Passenger-side pressure sensor internal malfunction	—	—	C, D	×
B1227:55	54		6	Configuration setting error (passenger-side pressure sensor structural malfunction)	—	—	C, D	×
[B1228:11]	35		16	Lap pre-tensioner seat belt circuit short to body ground	—	—	C, D	×
[B1228:12]				Lap pre-tensioner seat belt circuit short to power supply	—	—	C, D	×
[B1228:13]				Lap pre-tensioner seat belt circuit open circuit or resistance high	—	—	C, D	×
[B1228:19]				Short circuit to lap pre-tensioner seat belt and other air bag module circuits	—	—	C, D	×
[B1228:1A]				Lap pre-tensioner seat belt circuit resistance low	—	—	C, D	×
[B1228:55]	54		6	Configuration setting error (lap pre-tensioner seat belt structural malfunction)	—	—	C, D	×

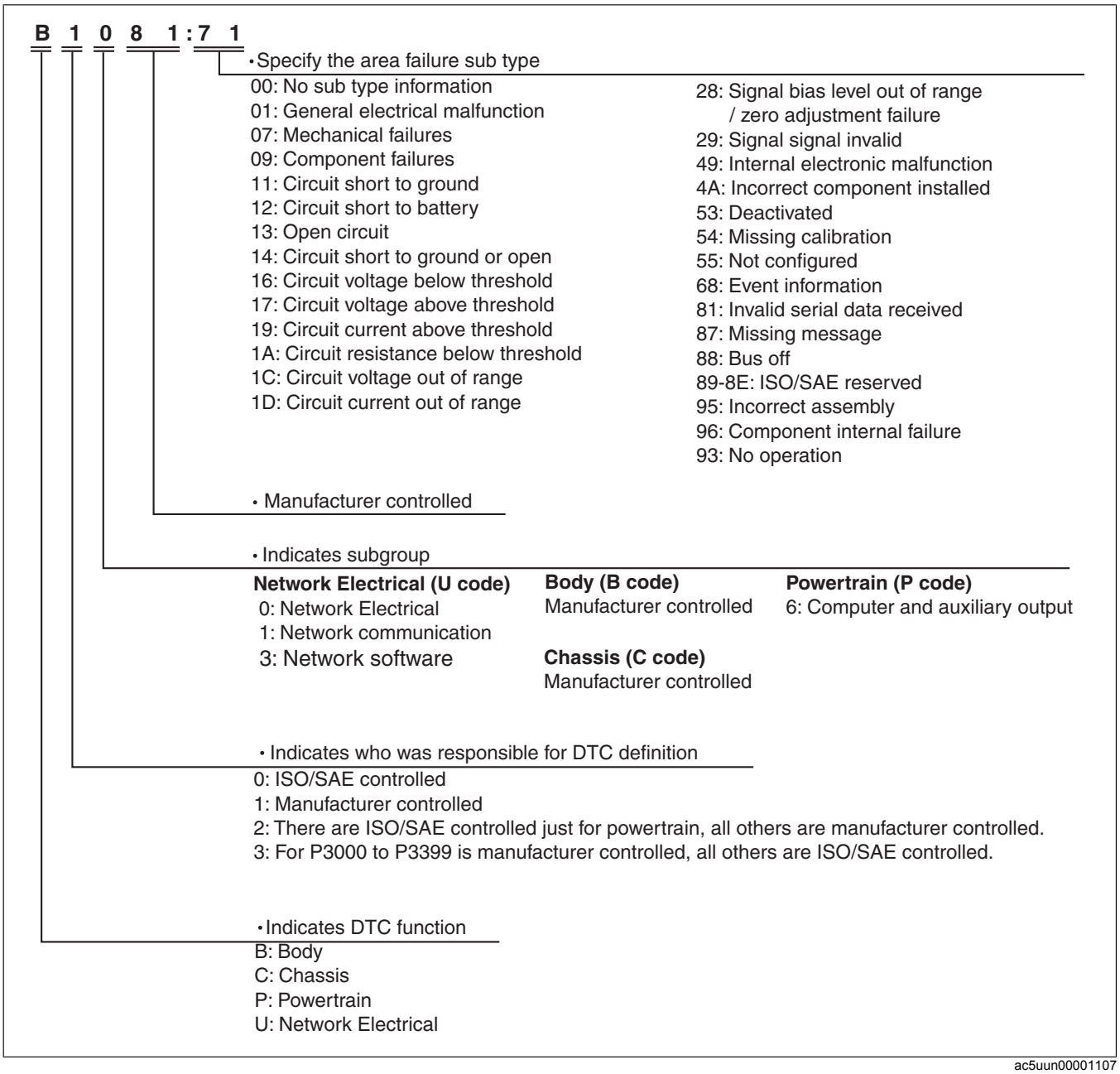
DTC				System malfunction location	Fail-safe function	Drive cycle	Self test type* 1	Memory function
M-MDS display	Air bag system warning light							
	Flashing pattern		Priority ranking					
B1410:11	33		24	Rear pre-tensioner seat belt (L.H.D.:LH/ R.H.D.:RH) circuit short to body ground	—	—	C, D	×
B1410:12				Rear pre-tensioner seat belt (L.H.D.:LH/ R.H.D.:RH) circuit short to power supply	—	—	C, D	×
B1410:13				Rear lap pre-tensioner seat belt (L.H.D.:LH/R.H.D.:RH) circuit open circuit or resistance high	—	—	C, D	×
B1410:19				Short circuit to rear pre-tensioner seat belt (L.H.D.:LH/R.H.D.:RH) and other air bag module circuits	—	—	C, D	×
B1410:1A				Rear pre-tensioner seat belt (L.H.D.:LH/ R.H.D.:RH) circuit resistance low	—	—	C, D	×
B1410:55	54		6	Configuration setting error (rear pre-tensioner seat belt (L.H.D.:LH/ R.H.D.:RH) structural malfunction)	—	—	C, D	×
B1412:11	32		23	Rear pre-tensioner seat belt (L.H.D.:RH/ R.H.D.:LH) circuit short to body ground	—	—	C, D	×
B1412:12				Rear pre-tensioner seat belt (L.H.D.:RH/ R.H.D.:LH) circuit short to power supply	—	—	C, D	×
B1412:13				Rear pre-tensioner seat belt (L.H.D.:RH/ R.H.D.:LH) circuit open circuit or resistance high	—	—	C, D	×
B1412:19				Short circuit to rear pre-tensioner seat belt (L.H.D.:RH/R.H.D.:LH) and other air bag module circuits	—	—	C, D	×
B1412:1A				Rear pre-tensioner seat belt (L.H.D.:RH/ R.H.D.:LH) circuit resistance low	—	—	C, D	×
B1412:55	54		6	Configuration setting error (rear pre-tensioner seat belt (L.H.D.:RH/ R.H.D.:LH) structural malfunction)	—	—	C, D	×
[B1417:11]	41		8	Passenger-side crash zone sensor circuit short to body ground	—	—	C, D	×
[B1417:12]				Passenger-side crash zone sensor circuit short to power supply	—	—	C, D	×
[B1417:29]				Passenger-side cash zone sensor internal malfunction	—	—	C, D	×
[B1417:2B]				Passenger-side crash zone sensor circuit short to other air bag sensor circuit	—	—	C, D	×
[B1417:4A]				Passenger-side crash zone sensor ID error	—	—	C, D	×
[B1417:87]				Signal reception error from passenger-side crash zone sensor	—	—	C, D	×
[B1417:96]				Passenger-side cash zone sensor internal malfunction	—	—	C, D	×
[B1417:55]	54		6	Configuration setting error (passenger-side crash zone sensor structural malfunction)	—	—	C, D	×
B143A:00	—	—	—	SAS control module operation (deployment) signal output	—	—	C, D	×
B1485:55	54		6	Configuration setting error (occupancy sensor structural malfunction)	—	—	C, D	×
B1D74:55	54		6	Configuration setting error (PAD switch structural malfunction)	—	—	C, D	×
B1D75:55	54		6	Configuration setting error (PAD switch structural malfunction)	—	—	C, D	×

DTC				System malfunction location	Fail-safe function	Drive cycle	Self test type* 1	Memory function
M-MDS display	Air bag system warning light							
	Flashing pattern		Priority ranking					
B1D75:62	56		27	PAD switch circuit open circuit or short to body ground	—	—	C, D	×
C0061:29	—	—	—	Low-G sensor (lateral-G) in SAS control module (internal circuit disabled)	—	—	C, D	×
C0062:29				Low-G sensor (forward-G) in SAS control module (internal circuit disabled)	—	—	C, D	×
C0063:29				Yaw rate sensor in SAS control module (internal circuit disabled)	—	—	C, D	×
U0001:88	—	Continuously illuminated	1	Module communication error (HS-CAN)	—	—	C, D	×
U0100:00	14		5	PCM Communication error or invalid value reception	—	—	C, D	×
U0155:00	14		5	Communication error with instrument cluster	—	—	C, D	×
U2100:00	—	Flashing	4	Configuration not set	—	—	C, D	×
U2107:00	—	—	—	SAS control module operation (deployment) signal output	—	—	C, D	×
U3000:49	—	Continuously illuminated	1	SAS control module internal malfunction	—	—	C, D	×
	12		2					
U3003:16	—	Continuously illuminated	1	SAS control module power supply voltage decreases (**less than 8 V**)	—	—	C, D	×
U3003:17				SAS control module power supply voltage increases (**16 V or more**)	—	—	C, D	×

*1 : C: CMDTC self test, D: ODDTC self test
[ ] : DTCs do not usually display because the related function is not equipped.

DTC 7-digit code definition

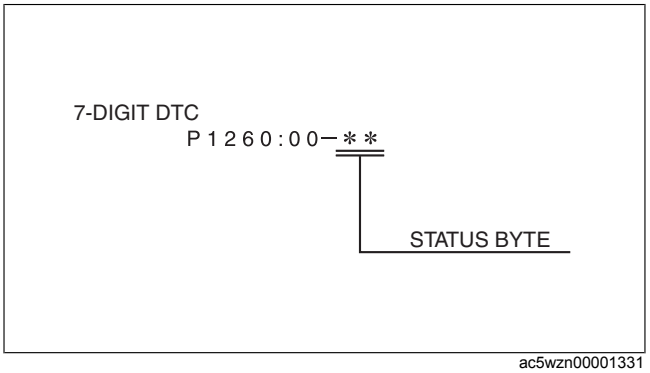
- When related systems or components have failed, the CM stores the DTC of the malfunctioning part in the CM memory, and allows for the retrieval of the stored data using the scanning tool when necessary. The DTCs are indicated using seven digits. Each digit indicates the following.



ac5uun00001107

Status byte for DTC

- The two digits (after hyphen) after the 7-digit DTC.
- A code which indicates the pending code, current/ past malfunction status, or warning illumination status.
- Can be read by performing a CMDTC self-test using the Mazda Modular Diagnostic System (M-MDS).
- For details on the status byte, refer to the explanation on the Mazda Modular Diagnostic System (M-MDS) when reading the DTC.



ac5wzn00001331

Snapshot data

Note

- The SAS control module stores the following two types of information when a DTC is detected and displays snap shot data in the Mazda Modular Diagnostic System (M-MDS).
  — Vehicle information detected by SAS control module

— Vehicle information detected by instrument cluster and received by SAS control module via CAN signal

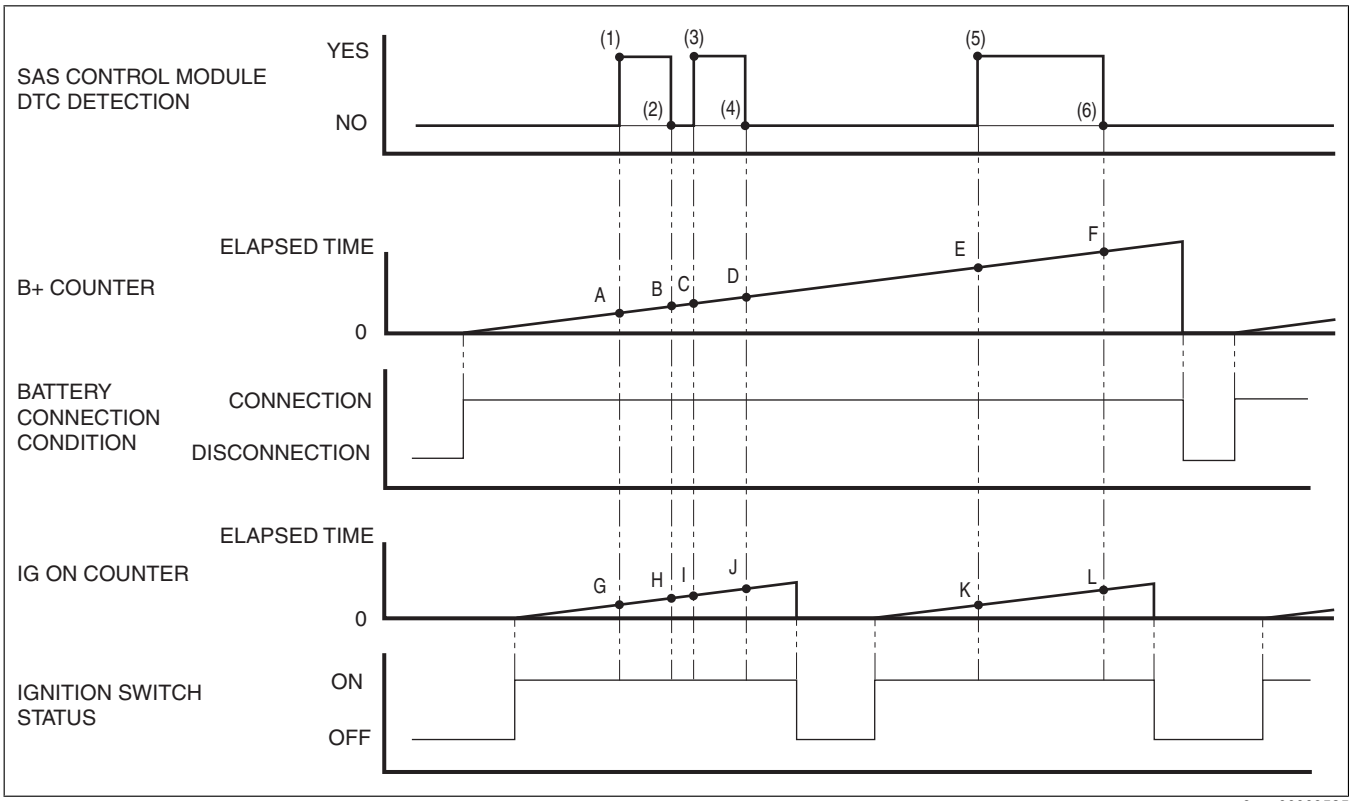
- The data for all DTCs currently detected is stored.

Snapshot data item	Unit	Definition	Data read/use method	Corresponding PID data monitor item
AAT	°C °F	Ambient temperature	—	—
APP_STATUS	Accelerator Pedal Off/ Under20%/ Over20%/ FAIL	Accelerator pedal position status	—	—
CFG_STATUS	Config Complete/ Not Configured/ Config Error	Instrument cluster configuration status	—	—
ECT_STATUS	Under 0 degrees C/ 0-Under 80 degrees C/ Over 80 degrees C/ FAIL	Engine coolant status	—	—
FAULT_CNT	—	Number of malfunction detections	—	—
FIRST_DET_IG	hh:mm:ss*1	Time when initial malfunction occurs (IG-ON counter) • Elapsed time from when ignition is switched ON (engine off or on) until SAS control module detects first malfunction*3	—	—
IC_VPWR	V	Instrument cluster power supply voltage	• The SAS control module constantly receives the power supply voltage value of the instrument cluster sent via CAN signal from the instrument cluster. • If a DTC is detected, the SAS control module records the power supply voltage of the instrument cluster when the DTC was detected, and it is displayed in the Mazda Modular Diagnostic System (M-MDS).	VPWR*2
IG-ON_TIMER	hh:mm:ss*1	Elapsed time since ignition was switched ON (engine off or on) **Note** • The instrument cluster records the elapsed time since the ignition was switched ON (engine off or on).	• The SAS control module constantly receives the elapsed time since the ignition was switched ON (engine off or on) sent via CAN signal from the instrument cluster. • If a DTC is detected, the SAS control module records the elapsed time since the ignition was switched ON (engine off or on) when the DTC was detected, and it is displayed in the Mazda Modular Diagnostic System (M-MDS).	—

Snapshot data item	Unit	Definition	Data read/use method	Corresponding PID data monitor item
LAST_CLR	hh:mm:ss*1	Repair time for last malfunction (B+ counter) • Elapsed time from when battery is connected until SAS control module detects that last malfunction is repaired*3	—	—
LAST_CLR_IG	hh:mm:ss*1	Repair time for last malfunction (IG-ON counter) • Elapsed time from when ignition is switched ON (engine off or on) until SAS control module detects that last malfunction is repaired*3	—	—
LAST_DET	hh:mm:ss*1	Time when last malfunction occurs (B+ counter) • Elapsed time from when battery is connected until SAS control module detects last malfunction*3	—	—
LAST_DET_IG	hh:mm:ss*1	Time when last malfunction occurs (IG-ON counter) • Elapsed time from when ignition is switched ON (engine off or on) until SAS control module detects last malfunction*3	—	—
PWR_MODE_KEY	Key Out/ Key Recently Out (Position 0)/ Accessory (Position 1)/ Post Ignition (Position 2)/ Ignition On (Position 2)/ Running (Position 2)/ Running - Starting	• Key Out: Ignition switched off • Key Recently Out (Position 0): Elapsed time **within 3 s** since ignition was switched off • Accessory (Position 1): Ignition is switched to ACC • Post Ignition (Position 2): Elapsed time **within 3 s** since ignition was switched ON (engine off or on) • Ignition On (Position 2): Ignition switched ON (engine off) • Running (Position 2): Ignition switched ON (engine on) • Running - Starting: Cranking condition	• The SAS control module constantly receives the ignition switch status sent via CAN signal from the instrument cluster. • If a DTC is detected, the SAS control module records the ignition switch status when the DTC was detected, and it is displayed in the Mazda Modular Diagnostic System (M-MDS).	—
RPM_STATUS	Engine Stop/ Under1500rpm/ Over1500rpm/ FAIL	Engine speed status	• The SAS control module constantly receives the engine speed sent via CAN signal from the instrument cluster. • If a DTC is detected, the SAS control module records the engine speed when the DTC was detected, and it is displayed in the Mazda Modular Diagnostic System (M-MDS).	TACHOMTR*2

Snapshot data item	Unit		Definition	Data read/use method	Corresponding PID data monitor item
SHIFT_STATUS	P/N/ D/ R/ FAIL		Selector lever position status	• The SAS control module constantly receives the selector lever position sent via CAN signal from the instrument cluster. • If a DTC is detected, the SAS control module records the selector lever position when the DTC was detected, and it is displayed in the Mazda Modular Diagnostic System (M-MDS).	—
TOTAL_DIST	km	miles	Accumulated total traveled distance from completion of vehicle until SAS control module detects DTC (Odometer value in instrument cluster)	The total traveled distance from which the SAS control module detects DTCs to the present can be calculated by performing the following procedure. 1. Verify the odometer value in the instrument cluster. 2. Verify the snapshot data item TOTAL_DIST. 3. Subtract 2 from 1.	—
TOTAL_TIME	hh:mm:ss*1		Accumulated total elapsed time since vehicle completion until SAS control module detects a DTC*3 **Note** • When the ROOM fuse is removed, and the ignition is switched off, the time is not included in the elapsed time.	The elapsed time from which the SAS control module detects DTCs to the present can be calculated by performing the following procedure. 1. Verify the instrument cluster PID item TOTAL_TIME. 2. Verify the snapshot data item TOTAL_TIME. 3. Subtract 2 from 1.	TOTAL_TIME*2
VPWR	V		SAS control module power supply malfunction	—	VPWR_IGA
VSPD_STATUS	Stop/ 0-10km/h/ Over10km/h/ FAIL		Vehicle speed status	• The SAS control module constantly receives the vehicle speed sent via CAN signal from the instrument cluster. • If a DTC is detected, the SAS control module records the vehicle speed when the DTC was detected, and it is displayed in the Mazda Modular Diagnostic System (M-MDS).	SPEEDOMTR*2

*1 : The seconds may be indicated after the decimal point.
*2 : Instrument cluster PID (See ON-BOARD DIAGNOSTIC [INSTRUMENT CLUSTER].)
*3 : The snap shot data of the number of stored malfunction detections, B+ counters, or IG ON counters is combined with the DTC detection conditions of the SAS control module, and the point-in-time data in the following table is stored.



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SAS control module DTC detection period	Snapshot data						
	FAULT_CNT (Number of malfunction detections)	**TOTAL_TIME**	**FIRST_DET_IG**	**LAST_DET**	**LAST_DET_IG**	**LAST_CLR**	**LAST_CLR_IG**
	Number of malfunction detections	**Time when first malfunction occurs (B+ counter)**	**Time when initial malfunction occurs (IG-ON counter)**	**Time when last malfunction occurs (B+ counter)**	**Time when last malfunction occurs (IG-ON counter)**	**Repair time for last malfunction (B+ counter)**	**Repair time for last malfunction (IG-ON counter)**
(1)	1	A	G	A	G	-	-
(2)	1	A	G	A	G	B	H
(3)	2	A	G	C	I	B	H
(4)	2	A	G	C	I	D	J
(5)	3	A	G	E	K	D	J
(6)	3	A	G	E	K	F	L

PID/Data monitoring function

- By using the PID/data monitoring function, the monitored item of the input/output signal, as set on the SAS control module, can be freely selected and read out in real-time.
- The Mazda Modular Diagnostic System (M-MDS) is used to read out PID/data monitor information.

PID	Unit/Condition	Data contents	SAS control module terminal
AB_WL (Air bag system warning light status)	Lamp_Off/ Lamp_On/ Plant_Mode/ Unknown	• Air bag system warning light off: Lamp_Off • Air bag system warning light on: Lamp_On • Air bag system warning light status: Plant_Mode • Air bag system warning light status: Unknown	3Q, 3S
D_SP_S_FLT (Seat track position sensor malfunction status)	**Note** • Displays in the M-MDS but it does not operate.		
D_SP_S_ST (Seat track position sensor status)	**Note** • Displays in the M-MDS but it does not operate.		

PID	Unit/Condition	Data contents	SAS control module terminal
OCS_DTC_CLR (Seat weight sensor DTC cleared status)	**Note** • Displays in the M-MDS but it does not operate.		
P_OCS_ST (Seat weight sensor status)	**Note** • Displays in the M-MDS but it does not operate.		
PAD_SW (Passenger air bag deactivation (PAD) switch status)	**Note** • Displays in the M-MDS but it does not operate.		
PAD_SW_FLT (Passenger air bag deactivation (PAD) switch malfunction status)	**Note** • Displays in the M-MDS but it does not operate.		
RES_C_AB_D (Driver-side curtain air bag module resistance nominal)	ohm	Continuous: 0.81–6.42 ohms	1W, 1U
RES_C_AB_P (Passenger-side curtain air bag module resistance nominal)	ohm	Continuous: 0.81–6.42 ohms	1AA, 1Z
RES_F_AB1_D (Driver-side air bag module (inflatorNo.1) resistance nominal)	ohm	Continuous: 0.99–6.42 ohms	3C, 3G
RES_F_AB1_P (Passenger-side air bag module (inflatorNo.1) resistance nominal)	ohm	Continuous: 0.81–6.42 ohms	3K, 3N
RES_F_AB2_D (Driver-side air bag module (inflatorNo.2) resistance nominal)	**Note** • Displays in the M-MDS but it does not operate.		
RES_F_AB2_P (Passenger-side air bag module (inflator No.2) resistance nominal)	**Note** • Displays in the M-MDS but it does not operate.		
RES_S_AB_D (Driver-side side air bag module resistance nominal)	ohm	Continuous: 0.81–9.85 ohms	1C, 1G
RES_S_AB_P (Passenger-side side air bag module resistance nominal)	ohm	Continuous: 0.81–9.85 ohms	1K, 1N
RES_SB_LP_D (Driver-side lap pre-tensioner seat belt resistance nominal)	**Note** • Displays in the M-MDS but it does not operate.		
RES_SB_LP_P (Passenger-side lap pre-tensioner seat belt resistance nominal)	**Note** • Displays in the M-MDS but it does not operate.		
RES_SB_P_D (Driver-side pre-tensioner seat belt resistance nominal)	ohm	Continuous: 0.81–6.42 ohms	2C, 2G

PID	Unit/Condition	Data contents	SAS control module terminal
RES_SB_P_P (Passenger-side pre-tensioner seat belt resistance nominal)	ohm	Continuous: 0.81–6.42 ohms	2K, 2N
RES_R_PR_D (Rear pre-tensioner seat belt status)	**Note** • Displays in the M-MDS but it does not operate.		
RES_R_PR_P (Rear pre-tensioner seat belt status)	**Note** • Displays in the M-MDS but it does not operate.		
PCD_A (Poorly connected detector bar terminal voltage nominal (SAS control module connector A))	mV	• Normal connected: 270.9 mV or less • Poor connected: 3173.4 mV or more	1a, 1b
PCD_B (Poorly connected detector bar terminal voltage nominal (SAS control module connector B))	mV	• Normal connected: 270.9 mV or less • Poor connected: 3173.4 mV or more	2a, 2b
PCD_C (Poorly connected detector bar terminal voltage nominal (SAS control module connector C))	mV	• Normal connected: 270.9 mV or less • Poor connected: 3173.4 mV or more	3a, 3b
SEAT_B_D (Driver-side buckle switch status)	Unbuckled/ Buckled	• Driver-side front seat belt fastened (Driver-side buckle switch off): Buckled • Driver-side front seat belt not fastened (Driver-side buckle switch on): Unbuckled	2U
SEAT_B_P (Passenger-side buckle switch status)	Unbuckled/ Buckled	• Passenger-side front seat belt fastened (Passenger-side buckle switch off): Buckled • Passenger-side front seat belt not fastened (Passenger-side buckle switch on): Unbuckled	2W
VPWR_IGA (SAS control module power voltage)	V	Ignition is switched ON: IG1 voltage	2AD